

SAFETY DATA SHEET

1. Identification of the substance /mixture and of the company/undertaking

Product Identifier Vitamin AD₃ 1000/200, feed grade
Product Code: 41000032
Recommended Use: Additive for animal nutrition in premixes and compound feeds

Manufacturer Information:

Company Name Zhejiang Medicine Co., Ltd. Changhai Biological Company
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2. Hazards identification

Hazard Classification

This product is not classified as a hazardous substance or as dangerous goods according to the classification

2.2 Other hazards

The product is under certain conditions capable of dust explosion.

3. Composition/information on ingredients

Chemical Name:	CAS No.	Concentration (w/w)*
Vitamin A Acetate	127-47-9	1,000,000IU/g (~ 40%)
Vitamin D ₃ Oil	67-97-0	200,000IU/g (~3%)
Gelatin	9000-70-8	~30%
Carbohydrates and Absorbents	-	21%~24%
Antioxidants	-	2%~5%
Anti-caking agent	-	~1%

Remarks: *The data is based on average.

4. First-aid measures

Description of first aid measures

Inhalation:	Move to fresh air. Consult physician immediately.
Skin contact:	Wash off immediately with plenty of water and soap. Seek medical advice if irritation persists.
Eye contact:	Flush eyes with plenty of water, also under the eyelids, for at least 15 minutes. Consult an eye specialist if necessary.
Ingestion:	Do not induce vomiting. Consult physician immediately.

Most important symptoms or effects, both acute and delayed

Symptoms: The most important known symptoms and effects are described in 2 and/or in 11. Further important symptoms and effects are so far not known.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-fighting measures

Extinguishing media

Suitable Extinguishing media: Water spray, foam, carbon dioxide (CO₂), dry powder.

Unsuitable Extinguishing media: None known.

Special hazards arising from the substance or mixture

Dust explosion hazard: fine dust dispersed in air in sufficient concentrations and in the presence of an ignition source is a potential dust explosion hazard.

Advice for fire-fighters

Wear full fire-fighting turn-out gear (full Bunker gear) including NIOSH-approved self-contained breathing apparatus.

6. Accidental release measures

Personal precautions

Avoid dust formation. Do not breathe dust. Ensure adequate ventilation. Use personal protective apparatus. Information regarding personal protective measures see 8.

Methods and material for containment and cleaning up

Sweep/shovel up or vacuum up. Collect in a suitable container for disposal. After cleaning, flush away traces with water.

7. Handling and storage

Precautions for safe handling - Provide for appropriate exhaust ventilation and dust

collection at machinery.

Conditions for safe storage

- Preparation may charge electricity: always use earth leads when transferring from one container to another.
- Store in a dry and well-ventilated place in the original container. Protected from air, heat and light. Keep at a temperature below 15°C to obtain higher retention rate of activities of Vitamin A and Vitamin D₃.

8. Exposure controls/personal protection

Personal protective measures

Respiratory protection:	Suitable respiratory protection like half-mask.
Hand protection:	Impervious gloves
Eye protection:	Safety glasses with side-shields (frame goggles)
Body protection:	Protective suit and Hygiene measures.

General safety and hygiene measures

When using, do not eat or drink. Remove and wash contaminated clothing before re-use. Ensure adequate ventilation, especially in confined areas.

9. Physical and chemical properties

Information on basic physical and chemical properties

Appearance:	Yellow to brown.
Form:	Micro-granulated
Odor:	Almost odorless
Odor threshold:	No data available
pH:	Not applicable
Flash point:	Not applicable
Evaporation rate:	Not applicable
Flammability (solid/gas):	May form combustible dust concentrations in air.
Vapour pressure:	Not applicable
Vapour density:	Not applicable
Bulk density:	0.55~0.70 g/cm ³
Solubility:	Insoluble in water
Partition coefficient: n-octanol/water:	No data available
Auto-ignition temperature:	Not applicable
Thermal Decomposition:	No data available
Viscosity:	Not applicable.

Explosion parameters

Maximum explosion pressure, $P_{\max}$ /MPa	0.48
Maximum rate of pressure rise (dp/dt), max(MPa/s)	5.7
Dust specific measure of explosibility, $K_{st}(Mpa \cdot m/s)$	1.5
Lowest explosion level, LEL(g/cm³)	50-100
Min Ignition temperature C (dust layer), MITL(°C)	> 400
Min Ignition temperature C (dust cloud), MITC(°C)	
Product particle size	480
Sieved sample, 100% <0.10mm	400
The minimum ignition energy, MIE/mJ	
Product particle size	250-500
Sieved sample, 100% <0.10mm	50-75

The MIE of a powder strongly depends on the particle size distribution and the water content of the tested sample: the finer and the dryer the powder the lower the MIE.

10. Stability and Reactivity

10.1 Reactivity:	No hazards to be specially mentioned.
10.2 Chemical stability:	Stable under recommended storage conditions.
10.3 Possibility of hazardous reactions:	Dust may form explosive mixture in air. (> 10 g/m ³)
10.4 Conditions to avoid:	Avoid all sources of ignition: heat, sparks, open flame, generating dust, moisture, static discharges and light.
10.5 Incompatible materials:	Incompatible with oxidizing agents, acid chlorides, acid anhydrides.
10.6 Hazardous combustion products:	No hazardous decomposition if used as directed.

11. Toxicological information**Information on toxicological effects**

<u>Acute toxicity:</u>	no data available observed
<u>Skin corrosion/ Irritation:</u>	no data available observed
<u>Eye damage/ irritation:</u>	no data available observed
<u>Sensitisation to the respiratory tract/skin:</u>	Non-sensitizing.
Carcinogenicity:	Not classified
Mutagenicity:	Not classified.
Reproductive toxicity:	May damage fertility or the unborn child.

STOT- single exposure:	Not classified.
STOT- repeated exposure:	Cause damage to organs through prolonged or repeated exposure.

12. Ecological information

Ecotoxicity:	No data available
Persistence and degradability:	Readily biodegradable.
Bioaccumulation potential:	No data available.
Mobility in soil:	No data available.
Results of PBT & vPvB assessment:	No data available.
Other adverse effects:	Harmful to aquatic life with long lasting effects.

13. Disposal considerations

Dispose of contents/container in accordance with local/regional/national/international regulations.

14. Transport information

Land transport

Not classified as a dangerous good under transport regulations

Sea transport

Not classified as a dangerous good under transport regulations

Air transport

Not classified as a dangerous good under transport regulations

15. Regulation Information

Safety, health and environmental regulations/legislation specific for the substance or mixture:

Relevant information regarding authorization: Not applicable.

Relevant information regarding restriction: Not applicable.

16. Other information

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